

Modeling of High-Speed DFB Lasers Considering the Spatial Holeburning Effect Using Three Rate Equations

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Abstract—A new approximated method of analyzing the effect of spatial holeburning on dynamic response of distributed-feedback (DFB) lasers is proposed. This method consists of a mode-stability analysis based on the coupled-wave theory and a dynamic analysis utilizing only three spatially independent rate equations. The rate equations can be solved using the calculated result of the stability analysis. The stability analysis predicts that the break of stable single-longitudinal mode (SLM) operation will occur before the modal threshold gain reaches the value for the next mode. Based on this method, calculated results of dc characteristics, large-signal responses, and AM responses, including harmonic distortion, are presented.

I. INTRODUCTION

SPATIAL holeburning in distributed-feedback (DFB) lasers was discussed for the first time by Soda *et al.* [1]. Since then, many works on both the static and the dynamic behaviors of DFB lasers considering the holeburning effect have been published [2]–[10]. The dynamic behavior of DFB lasers can be more clearly discussed using more exact calculation models such as CLADISS [5]. Such models utilize, particularly for analyzing large-signal responses, many differential equations corresponding to the segments divided along the laser cavity [5]–[10].

Strained/unstrained MQW structures are losing novelty because of recent great improvement in MOVPE growth technique. It is indispensable for high-speed DFB lasers to introduce a strained multiple-quantum-well (MQW) structure [11], although the carrier transport phenomenon was found to diminish high-speed performance [12]–[15]. The analysis of the high-speed phenomena in strained/unstrained MQW DFB lasers is extremely complicated because we have to solve simultaneously both the coupled-wave equation system and the rate equation system. Both equation systems have their own histories. Hence the simultaneous combination of both systems seems to give some difficulties to researchers who have been working primarily based on either equation system. It must be therefore important to introduce a simple analysis model in which both equation systems can be analyzed separately.

We propose in this paper a very simplified model for analyzing the holeburning effects on high-speed DFB lasers.

This model has been developed based on our previous work [2], in which the spatial dependence in the rate equations is eliminated by integrating them over the entire cavity. Only three rate equations are required to analyze the holeburning effects. In the previous work, however, we assumed sinusoidal photon profiles along the cavity and linear change of the lasing condition. The new approach presented here covers arbitrary profiles and self-consistent analysis of the lasing condition. We demonstrate that the dynamic mode instability problem can be analyzed separately from the rate equation analysis using only transfer matrix (*F*-matrix) method [1], [2]. The three rate equations are then easily solved using the results of the above mode-stability analysis. This simplifies the analysis of the large-signal responses. The dc-operated behavior is easily calculated by setting the time derivative terms in the rate equations zero. The small-signal AM responses, including harmonic distortion, are also calculated easily using the conventional algebraic method at the bias point analyzed in the dc analysis.

This paper is organized as follows. A theoretical formulation of the new method is explained in Section II. Examples of calculated results and discussion are demonstrated in Section III. Finally, a summary is shown in Section IV.

II. FORMULATION

A. Mode Stability Analysis Based on the Coupled-Wave Theory

In this subsection, the mechanism of spatial holeburning and the mode stability problem in DFB lasers are analyzed.

We assume photon density S and carrier density N along the cavity direction z as follows:

$$S(z, t) \equiv S(t)\{1 + f_1(z, t)\} \quad (1)$$

$$N(z, t) \equiv N(t)\{1 + f_2(z, t)\} \quad (2)$$

We have defined $f_{1,2}(z, t)$ as

$$\int_0^L f_{1,2}(z, t) dz = 0. \quad (3)$$

By this definition, $S(t)$ and $N(t)$ imply the average photon and carrier densities along z . Fig. 1 illustrates an example of the $S(z)$ and $N(z)$ profiles. The terms, $S(t)f_1(z, t)$ and $N(t)f_2(z, t)$, denote the spatial deviations from the averages, $S(t)$ and $N(t)$, respectively.

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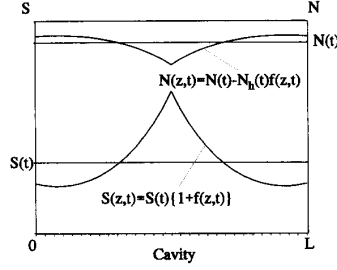


Fig. 1. Definition of nonuniformity functions (profiles of photon and carrier density along cavity direction Z).

A photon density distribution $S(t)f_1(z,t)$ usually exists in DFB lasers, depending on their specific structure. $S(t)f_1(z,t)$ produces the distribution $N(t)f_2(z,t)$. In dynamic analysis, in principle, we have to obtain self-consistent solutions of $f_1(z,t)$, $f_2(z,t)$, $S(t)$ and $N(t)$ using both the coupled-wave theory and rate equations.

Here we introduce a deviation function $f(z,t)$ common to both photon and carrier density distributions and a carrier density amplitude $N_h(t)$.

$$S(z,t) \equiv S(t)\{1 + f(z,t)\} \quad (4)$$

$$N(z,t) \equiv N(t) - N_h f(z,t). \quad (5)$$

That is, we employed an approximation of $N(t)f_2(z,t) \approx N_h(t)f_1(z,t) \equiv -N_h(t)f(z,t)$. We should introduce a z -dependent function $N_h(z,t)$ for strict transform of (2) into (5). The approximation of $N_h(z,t) \approx N_h(t)$ means that all the z -dependent differences between $S(z,t)$ and $N(z,t)$ are ignored. However, this approximation is considered to be valid if the z -dependent deviations from the average densities $S(t)$ and $N(t)$ are not extremely large.

The two primary mechanisms producing errors are predicted to be 1) large values of $S(t)$, which enhances the effect of the nonlinear gain term ϵ , and 2) spontaneous emission and/or nonradiative emission with a strong z -dependent effect.

The mechanism 1) will occur at high power levels, and mechanism 2) will occur at just above the threshold.

As long as the above approximation is allowed, we can carry out self-consistent analysis independently of the rate equation analysis, as explained later.

Let us proceed to the second stage of the holeburning phenomenon. The carrier density distribution $-N_h(t)f(z,t)$ gives rise to a refractive index change in the active region via dn/dN . This finally causes effective phase-shifting corresponding to the index distribution. The profile of the phase-shifting is given by the following equation in terms of the shift from the parameter δ , which denotes the deviation from the Bragg condition.

$$\begin{aligned} \Delta \delta(z,t) &= \frac{2\pi}{\lambda} \Gamma \frac{dn}{dN} N_h(t) f(z,t) \\ &= \alpha \frac{n_{eff}}{2c_0} g \Gamma N_h(t) f(z,t) \end{aligned} \quad (6)$$

where Γ is the mode confinement factor and λ is the lasing wavelength.

Here we can rewrite (6) into the following general form.

$$\Delta \delta(z,t) = \varphi(t) f(z,t). \quad (7)$$

We can thus analyze the mode stability problem mathematically unaffected by the detailed holeburning mechanism. Here $f(z,t)$ remains as the only z -dependent variable. To eliminate the z -dependence from the equations, squaring (7) and integrating all over z .

$$\Phi(t) = \phi(t) \sigma(t) \quad (8)$$

where

$$\begin{aligned} \Phi(t) &\equiv \frac{1}{L} \int_0^L \Delta \delta^2(z,t) dz, \\ \sigma(t) &\equiv \frac{1}{L} \int_0^L f^2(z,t) dz, \quad \phi(t) \equiv \varphi^2(t). \end{aligned} \quad (9)$$

The new change in the phase-shifting structure introduces new values of $f(z,t)$, internal mirror loss factor α_{th} , and the phase deviation δ . We therefore have to determine these values using the F -matrix formulation based on the coupled-wave theory [1], [2].

Defining $K(\Phi)$ as a function specifying a result of the above calculation step

$$\sigma(t + \Delta t) = K\{\Phi(t)\} = K\{\phi(t)\sigma(t)\} \quad (10)$$

Equations (8) and (10) specify a step of calculations for obtaining a solution of the nonuniformity variable $\sigma(t)$. If we can finally obtain self-consistent value of σ by the converging iterations of this step, the lasing longitudinal mode exists with a continuous change from the initial condition at threshold. If we fail to seek such a converged value, the mode no longer exists continuously. In other words, the mode becomes unstable chaotically if the calculation tends to be divergent.

The derivative expression of (10) is given by

$$\frac{\partial \sigma}{\partial t} = \frac{\partial K}{\partial \phi} \frac{\partial \phi}{\partial t} / \left(1 - \frac{\partial K}{\partial \sigma}\right). \quad (11)$$

The mode instability is considered to be related to the pole of (11).

If we can derive three z -independent rate equations containing $S(t)$, $N(t)$, $\phi(t)$ as a function of $N_h(t)$, and $\sigma(t)$, we can determine these four unknown quantities by combining (11) with the rate equations. However, the numerical values of $\partial K/\partial \phi$ and $\partial K/\partial \sigma$ should be obtained simultaneously using the F -matrix analysis. Noticing that (11) holds in general with respect to any variables other than t , we can rewrite it as follows.

$$\frac{\partial \sigma}{\partial \phi} = \frac{\partial K}{\partial \phi} / \left(1 - \frac{\partial K}{\partial \sigma}\right). \quad (12)$$

The above equation implies that the self-consistent value of σ can be determined uniquely for a fixed value of ϕ once a initial condition is given. That is, as a parameter of ϕ , self-consistent values of σ (accompanied by the values of α_{th} and δ) can be uniquely determined as long as the iterative calculation is convergent. Here the parameter ϕ can be interpreted as a generalized strength of spatial holeburning.

As a result, the rate equations can be solved using the predetermined numerical table of σ versus $\phi(N_h)$ as if they were independent of the F -matrix method.

The stability limit of ϕ must be a single-longitudinal-mode, (SLM) breakpoint. The DFB laser structure with larger ϕ_{break} is capable of more stable SLM operation. The SLM instability problem should be discussed from this viewpoint, not from the viewpoint of the threshold gain difference $\Delta\alpha L$. It is noteworthy that the SLM instability is likely to occur before $\Delta\alpha L$ reaches zero. The above-mentioned viewpoint is most important in this paper because such an idea has been never proposed as far as the author knows.

Equation (12) contains a feedback loop with regard to the nonuniformity variable σ . The instability of the mode is considered to occur due to the positive feedback from the nonuniformity itself. (Soda *et al.* carried out such iterative calculations in a similar manner at fixed values of normalized applied current, although they did not mention the mode instability before $\Delta\alpha L$ reaches zero. [1].)

B. Three z -independent Rate Equations, Considering Spatial Holeburning

In this subsection, three z -independent rate equations are derived.

Starting from the following fundamental rate equations

$$dS/dt = \Gamma g(N - N_0)(1 - \epsilon S)S - S/\tau_p + \Gamma \beta B N^2 \quad (13)$$

$$dN/dt = -g(N - N_0)(1 - \epsilon S)S - B N^2 + \eta_i I/eV \quad (14)$$

where g is differential gain, ϵ is gain compression factor, β is the fraction of spontaneous emission coupled into the lasing mode, τ_p is photon lifetime, B is the recombination coefficient, V is the active layer volume, e is electron charge, η_i is internal quantum efficiency, and I is injection current. S and N are photon and carrier densities in the active layer. N_0 is the carrier density for population inversion.

The radiant recombination term of $B N^2$ in (2) is introduced so as to be consistent with the parabolic approximation of threshold current given later by (22).

We may utilize various types of fundamental rate equations optionally, corresponding to the specific laser structure.

Substituting (4) and (5) into the fundamental rate in (13) and (14), and integrating with z over the entire cavity length L , we can obtain the following three z -independent rate equations.

$$\begin{aligned} dS/dt &= \Gamma g(N - N_0 - \sigma_2 N_h)S \\ &\quad - S/\tau_p - \epsilon \Gamma S^2 g[(N - N_0)(1 + \sigma_2) \\ &\quad - N_h(2\sigma_2 + \sigma_3)] + \Gamma \beta B(N^2 + \sigma_2 N_h^2) \\ dN/dt &= -g(N - N_0 - \sigma_2 N_h)S - B(N^2 + \sigma_2 N_h^2) \\ &\quad + \epsilon S^2 g[(N - N_0)(1 + \sigma_2) - N_h(2\sigma_2 + \sigma_3)] + \eta_i I/eV \end{aligned} \quad (15)$$

$$\begin{aligned} \frac{dN_h}{dt} + N_h \left(\frac{d\sigma_2}{dt} \right) / 2\sigma_2 &= g \left[N - N_0 - N_h \left(1 + \frac{\sigma_3}{\sigma_2} \right) \right] S \\ &\quad - B N_h \left(2N - N_h \frac{\sigma_3}{\sigma_2} \right) - \epsilon S^2 g \left[(N - N_0) \left(2 + \frac{\sigma_3}{\sigma_2} \right) \right. \\ &\quad \left. - N_h \left(1 + 2 \frac{\sigma_3}{\sigma_2} + \frac{\sigma_4}{\sigma_2} \right) \right] \end{aligned} \quad (17)$$

where

$$\begin{aligned} \sigma_2 &\equiv \frac{1}{L} \int_0^L f^2(z, t) dz, \\ \sigma_3 &\equiv \frac{1}{L} \int_0^L f^3(z, t) dz, \quad \sigma_4 \equiv \frac{1}{L} \int_0^L f^4(z, t) dz. \end{aligned} \quad (18)$$

To obtain (17), we multiply (14) by $f(z, t)$ before integrating it.

The inverse of photon life time τ_p is expressed as follows.

$$\tau_p^{-1}(t) = \frac{c_0}{n_{eff}} [\Gamma \alpha_{ac} + (1 - \Gamma) \alpha_{ex} + 2 \alpha_{th}(t)] \quad (19)$$

where, c_0 is the speed of light in vacuum, n_{eff} is effective refractive index of waveguide structure, α_{ac} is optical loss factor of the active layer, α_{ex} is optical loss factor outside the active region, and α_{th} is mirror-loss factor calculated from the coupled-wave analysis for DFB lasers. The change in photon life time arises from the α_{th} change.

The z -independent parameters, $\sigma_{2,3,4}$ and τ_p represent the nonuniformity along z . The numerical tables for these four parameters can be obtained beforehand in a manner similar to that in the previous subsection. We can thus consider the holeburning effects using the above three rate equations. The most important parameter is the squared deviation, σ_2 the same parameter as in the previous subsection. This also agrees with the parameter F given on p. 812 in [1], which Soda *et al.* introduced as an index for the strength of spatial holeburning. If we assume a sinusoidal distribution along z as in our previous work, the cubic deviation σ_3 equals zero. The fourth-powered deviation σ_4 affects the saturation behavior of N_h .

The third rate equation (17) has been sometimes approximated to be time-independent, as in the analysis for transverse-holeburning [16]. That is, all the effects of holeburning are incorporated into some constants adding to ϵ . However, the derivative term, dN_h/dt , cannot be neglected in the axial holeburning analysis because the cavity length is long enough to ignore carrier diffusion effect.

The rate equation model presented here has the following features:

- 1) All the variables can be written as if spatially independent.
- 2) The carrier density component of spatial nonuniformity N_h can be considered separately from the average carrier density component N , which is familiar to us.
- 3) The three rate equations could be solved as if they were independent of the coupled-mode analysis using the calculated numerical table for the σ -parameters.

Feature 3) is of striking advantage to simplifying the analysis considering spatial holeburning.

Fig. 2 shows the difference between this model and the large-scale models. The large-scale models require full-linkage between $m \times 2$ rate equations and m F -matrices, time by time. The model presented here, on the contrary, requires only three spatially independent rate equations and four variables expressed as a function of N_h . In other words, the two analysis systems, rate equations, and F -matrix, are linked by N_h . Self-consistency is guaranteed throughout the analysis. If the

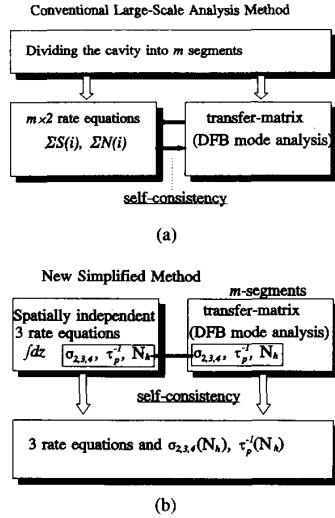


Fig. 2. Chart for comparing the model presented here and the large-scale models.

self-consistent F -matrix calculation does not converge, this mode is considered to be no longer permitted to oscillate continuously, hopping to the other modes with self-consistent solutions.

The power $P(t)$ from the output facet at $z = 0$ is written in terms of the average photon density $S(t)$ as follows.

$$P(t) = h\nu \frac{c_0}{n_{eff}} 2\alpha_{th}(t)L \cdot S(t) \frac{dW}{\Gamma} \cdot \frac{\{1 + f(0, t)\}(1 - R_f)}{\{1 + f(0, t)\}(1 - R_f) + \{1 + f(L, t)\}(1 - R_r)} \quad (20)$$

where d and W are the thickness and width of the active layer ($V = dWL$), and R_f and R_r are the reflectivities at front and rear facets, respectively.

Threshold carrier density can be obtained by assuming $dN/dt = 0$, $dS/dt = 0$, $N_h = 0$, and $S = 0$, in (15) and (16). (We assume $\Gamma g(N_{th} - N_0) - 1/\tau_p = 0$ in (15), and $\epsilon = 0$).

$$N_{th} = (\Gamma g \tau_p)^{-1} + N_0 = (\eta_i I_{th} / eVB)^{1/2}. \quad (21)$$

Threshold current I_{th} is expressed as follows from the above equation [17].

$$I_{th} = (edWLBc_0^2/n_{eff}^2 \eta_i) \cdot \left[\frac{n_{eff} N_0}{c_0} + \frac{\alpha_{ac}}{g} + \frac{(1 - \Gamma)\alpha_{ex}}{g\Gamma} + \frac{2\alpha_{th}}{g\Gamma} \right]^2. \quad (22)$$

C. dc Analysis

The dc behavior can be easily analyzed by setting the time-derivative terms in the rate equations (15), (16), and (17), and the spontaneous β term to be zero. The N_h -dependent variables, $\sigma_{2,3,4}$ and τ_p^{-1} have been already obtained by the F -matrix analysis for specific DFB laser structures defined by coupling coefficient κL , facet reflectivity including phase, and phase-shift etc.

D. Large-Signal Analysis

Large-signal responses can be analyzed using the rate equations (15), (16) and (17). The functions, $\sigma_{2,3,4}(N_h)$ and $\tau_p^{-1}(N_h)$ have been predetermined by the F -matrix analysis.

The time-dependent lasing wavelength shift (chirp) in the absence of thermal effect is expressed by the following equation, as in our previous work [2].

$$\Delta \lambda(t) = \frac{\lambda}{n_{eff}} \frac{dn}{dN} \Gamma [N(t) - N(0)] - \frac{\lambda^2}{2\pi n_{eff}} [\delta(t) - \delta(0)]. \quad (23)$$

The first term of the above equation denotes the shift of the Bragg wavelength due to the variation of the average carrier density. The change in photon lifetime due to spatial holeburning is implicitly included in this term. The second term implies the shift of the lasing mode within the stop-band as a result of spatial holeburning.

We can rewrite (23) using the α -parameter as follows.

$$\Delta \lambda(t) = \frac{4\pi \alpha}{c_0} g \Gamma [N(t) - N(0)] - \frac{\lambda^2}{2\pi n_{eff}} [\delta(t) - \delta(0)]. \quad (24)$$

The second term of the above equation is also related implicitly to the α -parameter via (6).

D. Small-Signal Analysis

We can analyze small-signal AM responses using the usual harmonic formulation and assuming the variables in the rate equations as follows

$$\begin{aligned} S(t) &\equiv S + s_1 e^{j\omega t} + s_2 e^{j2\omega t}, \\ N(t) &\equiv N + n_1 e^{j\omega t} + n_2 e^{j2\omega t}, \\ N_h(t) &\equiv N_h + n_{h1} e^{j\omega t} + n_{h2} e^{j2\omega t}, \\ I(t) &\equiv I + i_1 e^{j\omega t} \end{aligned} \quad (25)$$

where S, N, N_h , and I are constants at a bias point for modulation.

Recalling that the rest of the variables in the rate equations, $\sigma_{2,3,4}$ and τ_p^{-1} were previously determined as functions of N_h , they can be expressed as follows using a Taylor expansion around a value of N_h at a bias point.

$$\begin{aligned} \sigma_m(N_h + \Delta N_h) &\simeq \sigma_m(N_h) + r_{m1} \Delta N_h \\ &\quad + \frac{1}{2} r_{m2} \Delta N_h^2 + \dots \quad (m = 2, 3, 4) \\ \tau_p^{-1}(N_h + \Delta N_h) &\simeq \tau_p^{-1}(N_h) + r_{11} \Delta N_h \\ &\quad + \frac{1}{2} r_{12} \Delta N_h^2 + \dots \end{aligned} \quad (26)$$

where

$$\begin{aligned} r_{m1} &\equiv \left. \frac{\partial \sigma_m}{\partial N_h} \right|_{N_h}, & r_{m2} &\equiv \left. \frac{\partial^2 \sigma_m}{\partial N_h^2} \right|_{N_h}, \\ r_{11} &\equiv \left. \frac{\partial \tau_p^{-1}}{\partial N_h} \right|_{N_h}, & r_{12} &\equiv \left. \frac{\partial^2 \tau_p^{-1}}{\partial N_h^2} \right|_{N_h} \end{aligned} \quad (27)$$

The harmonic expressions for $\sigma_{2,3,4}(t)$ and $\tau_p^{-1}(t)$ become as follows by substituting $\Delta N_h \equiv n_{h1} e^{j\omega t} + n_{h2} e^{j2\omega t}$

into (26)

$$\begin{aligned}\tau_p^{-1} &\equiv \tau_p^{-1} + r_{11}n_{h1}e^{j\omega t} + (r_{11}n_{h2} + r_{12}n_{h1}^2/2)e^{2j\omega t} \\ \sigma_2(t) &\equiv \sigma_2 + r_{21}n_{h1}e^{j\omega t} + (r_{21}n_{h2} + r_{22}n_{h1}^2/2)e^{2j\omega t} \\ \sigma_3(t) &\equiv \sigma_3 + r_{31}n_{h1}e^{j\omega t} + (r_{31}n_{h2} + r_{32}n_{h1}^2/2)e^{2j\omega t} \\ \sigma_4(t) &\equiv \sigma_4 + r_{41}n_{h1}e^{j\omega t} + (r_{41}n_{h2} + r_{42}n_{h1}^2/2)e^{2j\omega t}.\end{aligned}\quad (28)$$

Substituting (25) and (28) into the rate equations (15), (16), and (17), and collecting separately terms with $\exp(j\omega t)$ and $\exp(2j\omega t)$ dependencies, the following six simultaneous complex equations are obtained. (The spontaneous term is neglected.)

$$\begin{aligned}(-j\omega + D_{11})s_1 + D_{12}n_1 + D_{13}n_{h1} &= 0 \\ D_{21}s_1 + (-j\omega + D_{22})n_1 + D_{23}n_{h1} &= -i_1/eV \\ D_{31}s_1 + D_{32}n_1 + [-j\omega(\sigma_2 + r_{21}N_h/2) + D_{33}]n_{h1} &= 0\end{aligned}\quad (29)$$

$$\begin{aligned}(-2j\omega + D_{11})s_2 + D_{12}n_2 + D_{13}n_{h2} &= P_1 \\ D_{21}s_2 + (-2j\omega + D_{22})n_2 + D_{23}n_{h2} &= P_2 \\ D_{31}s_2 + D_{32}n_2 + [-2j\omega(\sigma_2 + r_{21}N_h/2) + D_{33}]n_{h2} &= P_3 - j\omega Q_3\end{aligned}\quad (30)$$

where

$$\begin{aligned}D_{11} &\equiv -\Gamma\epsilon g S[(N - N_0)(1 + \sigma_2) - N_h(2\sigma_2 + \sigma_3)] \\ D_{12} &\equiv \Gamma g S[1 - \epsilon S(1 + \sigma_2)] \\ D_{13} &\equiv -\Gamma g S[\sigma_2 + r_{21}N_h - \epsilon S\{(2\sigma_2 + \sigma_3) \\ &\quad - r_{21}(N - N_0) + (r_{21} + r_{31})N_h\}] - r_{11}S \\ D_{21} &\equiv -g(N - N_0 - \sigma_2N_h) + 2\epsilon g S[(N - N_0)(1 + \sigma_2) \\ &\quad - N_h(2\sigma_2 + \sigma_3)] \\ D_{22} &\equiv -2BN - gS[1 - \epsilon S(1 + \sigma_2)] \\ D_{23} &\equiv gS[\sigma_2 + r_{21}N_h - \epsilon S\{(2\sigma_2 + \sigma_3) - r_{21}(N - N_0) \\ &\quad + (r_{21} + r_{31})N_h\}] - BN_h(2\sigma_2 + r_{21}N_h) \\ D_{31} &\equiv g[(N - N_0 - N_h)\sigma_2 - N_h\sigma_3] \\ &\quad - 2\epsilon g S[(N - N_0)(2\sigma_2 + \sigma_3) - N_h(\sigma_2 + 2\sigma_3 + \sigma_4)] \\ D_{32} &\equiv -2B\sigma_2N_h + gS[\sigma_2 - \epsilon S(2\sigma_2 + \sigma_3)] \\ D_{33} &\equiv -2B(\sigma_2 + r_{21}N_h)N + B(2\sigma_3 + r_{31}N_h)N_h \\ &\quad + gS[-(\sigma_2 + \sigma_3) + r_{21}(N - N_0 - N_h) - r_{31}N_h] \\ &\quad - \epsilon g S^2[-\sigma_2 - 2\sigma_3 - \sigma_4 + (N - N_0)(2r_{21} + r_{31}) \\ &\quad - N_h(r_{21} + 2r_{31} + r_{41})] \\ P_1 &\equiv -\Gamma g[s_1n_1 - (r_{21} + r_{22}N_h/2)Sn_{h1}^2 \\ &\quad - (\sigma_2 + r_{21}N_h)n_{h1}s_1] \\ &\quad + \epsilon \Gamma g[\{(1 + \sigma_2)(N - N_0) - (2\sigma_2 + \sigma_3)N_h\}s_1^2 \\ &\quad + 2Ss_1\{(1 + \sigma_2)n_1 - (2\sigma_2 + \sigma_3)n_{h1} \\ &\quad - r_{21}(N - N_0)n_{h1} - (r_{21} + r_{31})N_hn_{h1}\} \\ &\quad - S^2n_{h1}^2(r_{21} + r_{31} + r_{22}N_h/2 + r_{32}N_h/2) \\ &\quad + S^2\{r_{21}n_1n_{h1} + r_{22}(N - N_0)n_{h1}^2/2\} \\ &\quad - r_{11}s_1n_{h1} - r_{12}Sn_{h1}^2/2] \\ P_2 &\equiv g[s_1n_1 - (r_{21} + r_{22}N_h/2)Sn_{h1}^2 - (\sigma_2 + r_{21}N_h)n_{h1}s_1] \\ &\quad - \epsilon g[\{(1 + \sigma_2)(N - N_0) - (2\sigma_2 + \sigma_3)N_h\}s_1^2\end{aligned}\quad (31)$$

$$\begin{aligned}&+ 2Ss_1\{(1 + \sigma_2)n_1 - (2\sigma_2 + \sigma_3)n_{h1} \\ &\quad - r_{21}(N - N_0)n_{h1} - (r_{21} + r_{31})N_hn_{h1}\} \\ &\quad - S^2n_{h1}^2(r_{21} + r_{31} + r_{22}N_h/2 + r_{32}N_h/2) \\ &\quad + S^2\{r_{21}n_1n_{h1} + r_{22}(N - N_0)n_{h1}^2/2\} \\ &\quad + Bn_{h1}^2(\sigma_2 + r_{21}N_h + r_{22}N_h^2/2) + Bn_{h1}^2 \\ P_3 &\equiv 2B[(\sigma_2 + r_{21}N_h)n_1n_{h1} + (r_{21} + r_{22}N_h/2)n_{h1}^2] \\ &\quad - B(\sigma_3 + 2r_{21}N_h + r_{32}N_h^2/2)n_{h1}^2 \\ &\quad - gs_1[\sigma_2n_1 + \{\sigma_2 + r_{21}(N - N_0 - N_h)\}n_{h1} \\ &\quad - (\sigma_3 + r_{31}N_h)n_{h1}] \\ &\quad - gS[r_{21}n_1n_{h1} - \{r_{21} + r_{31} - (N - N_0 - N_h)r_{22}/2 \\ &\quad + N_hr_{32}/2\}n_{h1}^2] \\ &\quad + \epsilon gs_1^2[(2\sigma_2 + \sigma_3)(N - N_0) - (\sigma_2 + 2\sigma_3 + \sigma_4)N_h] \\ &\quad - \epsilon gS^2[(2r_{21} + r_{31})n_1n_{h1} + (N - N_0)\frac{2r_{22} + r_{32}}{2}n_{h1}^2 \\ &\quad - (r_{21} + 2r_{31} + r_{41} + \frac{r_{22} + 2r_{32} + r_{42}}{2}N_h)n_{h1}^2] \\ &\quad - 2\epsilon gSs_1[(N - N_0)(2r_{21} + r_{31})n_{h1} + 2(\sigma_2 + \sigma_3)n_1 \\ &\quad - \{(\sigma_2 + 2\sigma_3 + \sigma_4) + (r_{21} + 2r_{31} + r_{41})N_h\}n_{h1}] \\ Q_3 &\equiv (3r_{21}/2 + r_{22}N_h/2)n_{h1}^2\end{aligned}\quad (32)$$

The values of $S, N, N_h, \sigma_{2,3,4}, \tau_p$ and r -parameters have been calculated as a result of the dc analysis at the bias current for modulation.

The simultaneous equations (30) can be solved by determining the P_1, P_2 , and Q_3 values from the solutions of (29), s_1, n_1 , and n_{h1} . Thus the small-signal frequency response $10 \times \log_{10} |s_1(\omega)/s_1(0)|$ and second harmonic distortion (2HD) $10 \times \log_{10} |s_2(\omega)/s_1(\omega)|$ can be calculated algebraically with respect to frequency.

E. Simplified Small-Signal Analysis

The holeburning phenomenon consists of two stages. The first stage means the occurrence of spatially nonuniform distributions of photon and carrier densities. The second means the nonlinearity caused by the changes of $\sigma_{2,3,4}$ and τ_p^{-1} . The latter can be observed as a nonlinearity in current–light output characteristics (I – P curve), emphasized by its derivative, dP/dI curve. The former, however, cannot be explicitly observed in usual measurements.

Let us try to simplify the rate equations to extract the influences of the nonuniformity itself at the first stage on the small-signal response. The terms incorporating σ_3 are assumed to be zero. The ϵ -terms are also neglected. Then only $\sigma_2(\equiv \sigma)$ remains among the σ -parameters. The time-derivatives, $d\sigma/dt$ and $d\tau_p^{-1}/dt$ are neglected to eliminate the effects of the second stage of spatial holeburning. Then we can pay attention only to σ and N_h in the analysis in this subsection. This approximation may be applied to DFB lasers with a small Γ value, such as strained MQW DFB lasers, which tend to have relatively linear I – P curves in spite of relatively strong nonuniformity.

The above approximation yields the following simplified rate equations.

$$dS/dt = \Gamma g(N - N_0 - \sigma N_h)S - S/\tau_p \quad (33)$$

$$\frac{s_1(\omega)}{s_1(0)} = \frac{\omega_r^2[1 - \sigma + \Delta\sigma^2(1 - \tau_n^{-1}\gamma^{-1})][1 + j\omega\{\gamma(1 - \sigma) + \sigma\tau_n^{-1}\}^{-1}]}{[1 - \sigma + \Delta\sigma^2(1 - \tau_n^{-1}\gamma^{-1})]\omega_r^2 - 2\omega^2 + j\omega[(1 - \sigma)(\omega_r^2 + \gamma^2) + \sigma\tau_n^{-2} - \omega^2]\gamma^{-1}} \quad (36)$$

$$\frac{s_2(\omega)}{s_1(\omega)} = \frac{M}{2}(1 - \sigma\Delta)[- \omega_r^2\sigma(\Delta\gamma + \tau_n^{-1}) - \omega^2\{\gamma + \tau_n^{-1}(1 - 2\sigma)\} + j\omega\{-\omega_r^2\sigma(\Delta + 2) - \gamma\tau_n^{-1}(1 - 2\sigma) - \omega^2(1 - \sigma)\}]$$

$$\cdot \frac{\omega_r^2(1 + j2\omega[\gamma(1 - \sigma) + \sigma\tau_n^{-1}]^{-1})}{[(1 - \sigma + \Delta\sigma^2(1 - 2\tau_n^{-1}\gamma^{-1}))\omega_r^2 - 8\omega^2 + j2\omega[(1 - \sigma)(\omega_r^2 + \gamma^2) + \sigma\tau_n^{-2} - 4\omega^2]\gamma^{-1}]}$$

$$\cdot \frac{1}{[1 - \sigma + \Delta\sigma^2(1 - 2\tau_n^{-1}\gamma^{-1})]\omega_r^2 - 2\omega^2 + j\omega[(1 - \sigma)(\omega_r^2 + \gamma^2) + \sigma\tau_n^{-2} - \omega^2]\gamma^{-1}} \cdot \frac{\tau_n^{-1} + j2\omega}{\tau_n^{-1} + j\omega} \quad (39)$$

$$dN/dt = -g(N - N_0 - \sigma N_h)S - N/\tau_n + I/eV \quad (34)$$

$$dN_h/dt = -g(N - N_0 - N_h)S - N_h/\tau_n \quad (35)$$

where carrier lifetime τ_n is used instead of the recombination coefficient B for further simplicity.

Substituting (25) into the above simplified rate equations and collecting separately terms with $\exp(j\omega t)$ and $\exp(2j\omega t)$ dependencies, the following AM response formula is obtained as (36) (see top of page) where

$$\omega_r^2 = gS/\tau_p, \quad \gamma = gS + \tau_n^{-1},$$

$$\Delta = \frac{N_h}{N - N_0} + \frac{\Gamma gS}{\Gamma gS + \tau_n^{-1}}. \quad (37)$$

In the absence of holeburning ($\Delta = \sigma = 0$), (36) becomes a familiar form of

$$\frac{s_1(\omega)}{s_1(0)} = \frac{\omega_r^2}{\omega_r^2 - \omega^2 + j\omega\gamma}. \quad (38)$$

Similarly, a formula for frequency response of 2HD is given by (39) (see top of page) where M is modulation index.

In the absence of holeburning ($\Delta = \sigma = 0$), the above equation becomes a familiar form of

$$\frac{s_2(\omega)}{s_1(\omega)} = \frac{M}{2} \frac{\omega_r^2}{\omega_r^2 - \omega^2 + j\omega\gamma} \cdot \frac{j\omega(\tau_n^{-1} + j2\omega)}{\omega_r^2 - 4\omega^2 + j2\omega\gamma} \quad (40)$$

From the above formulas, the AM response is considered to be affected by the nonuniformity itself expressed by σ and N_h .

III. CALCULATED RESULTS AND DISCUSSION

The calculated results of the mode-stability analysis, dc characteristics, large-signal responses, and small-signal AM responses, including 2HD, are shown and discussed in this section.

The calculations presented here are all carried out using the values listed in Table I, assuming $1.3 \mu\text{m}$ -region bulk $\lambda/4$ phase-shift DFB lasers with zero reflectivity facets. The cavity is divided into 30 segments for the F -matrix calculations. The computing programs used here are written in FORTRAN-77. The Lunge-Kutta-Verner method is used for solving the rate equations, and Müller's method is used for solving the complex equations in the F -matrix analysis.

The values of the nonuniformity parameters, $\sigma_{2,3,4}$ at threshold are calculated for the $\lambda/4$ phase-shift DFB lasers as a function of κL , as shown in Fig. 3. This will be useful for

TABLE I
PARAMETER LIST FOR CALCULATION

parameter	description	value	unit
g	differential gain	$2.5 \cdot 10^{-6}$	cm^3s^{-1}
Γ	optical confinement factor	0.2408	
ϵ	gain compression factor	$2 \cdot 10^{-17}$	cm^3
B	recombination coefficient	$8 \cdot 10^{-11}$	cm^3s^{-1}
L	cavity length	300	μm
W	width of active region	1	μm
d	thickness of active region	0.12	μm
α_{ac}	optical loss factor in the active region	100	cm^{-1}
α_{ex}	optical loss outside the active region	10	cm^{-1}
n_{eff}	effective refractive index of guiding structure	3.2512	
S	average photon density		cm^{-3}
N	average carrier density		cm^{-3}
N_h	amplitude of carrier density distribution along z		cm^{-3}
N_0	carrier density for population inversion	$2 \cdot 10^{18}$	cm^3
β	spontaneous emission fraction for lasing	10^{-3}	
$\sigma_{2,3,4}$	holeburning factors		
η_i	internal quantum efficiency	0.8	
dn/dN	change rate of refractive index	$-2.0 \cdot 10^{-20}$	cm^3
λ	lasing wavelength	1.3	μm
e	charge of an electron	$1.60219 \cdot 10^{-19}$	C

estimating the initial strength of the holeburning effects. As in Fig. 3, these parameters are very small if $1 < \kappa L < 2$. The cubic deviation σ_3 is smaller than σ_2 in a wide range of

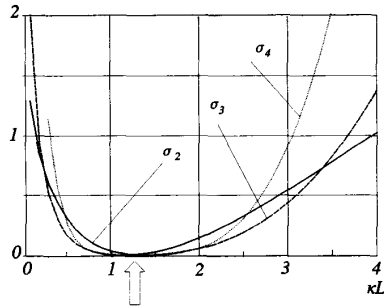


Fig. 3. Calculated results of basic holeburning parameters, $\sigma_{2,3,4}$, for perfectly AR-coated $\lambda/4$ phase-shift DFB lasers as a function of κL .

$0.3 < \kappa L < 3.3$. The parameter σ_4 changes most rapidly for $\kappa L < 0.6$ and $\kappa L > 2.5$.

A. Mode Stability Analysis

We carry out the self-consistent calculations at fixed values of N_h , instead of applied current as in [1]. The values of $\sigma_{2,3,4}$ and τ_p^{-1} in the rate equations can be expressed as a function of N_h , $\sigma_{2,3,4}(N_h)$ and $\tau_p^{-1}(N_h)$. We can also express as $\alpha_{th} L(N_h)$, $\delta L(N_h)$, and $\Delta \alpha L(N_h)$, where $\Delta \alpha L$ is threshold gain difference from the next mode.

Figs. 4(a) and (b) show the plots of the squared deviation σ_2 and the threshold gain difference $\Delta \alpha L$ as a function of $\phi = -\Gamma(2\pi/\lambda)(dn/dN)LN_h$, respectively. They are plotted for the $\lambda/4$ phase-shift DFB lasers with $\kappa L = 1, 1.3, 2$, and 3 . A normalized variable ϕ is utilized instead of N_h . We start the calculation from the initial profile at threshold. Then the steps of the iterative self-consistent calculations are repeated by a period of $\phi = 0.05\pi$. The calculation for a new step is initiated from the converged value obtained at the previous step. The calculated results do not depend on the period of ϕ . Examples of the illustrated traces of the iterative calculations at typical three points on the curve for $\kappa L = 2$ are inserted into Fig. 4(a). The numerical convergence on a self-consistent value is much faster for smaller values of $|\phi|$. However, the number of iterations required for convergence rapidly increases when ϕ approaches a point denoted by open circles. We cannot obtain self-consistent solutions if $|\phi|$ exceeds this point, implying that a stable longitudinal mode no longer exist continuously. It should be noted that the gain difference $\Delta \alpha L$ does not yet reach zero at the instability points. A mode hopping to the other stable mode is then presumed to occur. The trend of numerically slow convergence just before the mode hopping might be related to the mode-hopping noise in DFB lasers.

The results in Fig. 4(b) almost agree with Fig. 15 in [1] obtained by Soda *et al.*, except that the abscissa in their Fig. 15 is scaled with normalized current density, $-\Gamma(dn/dN)N_{th}(J_c/J_{th} - 1)$.

Examples of two different types of dP/dI curves are depicted in Fig. 5(a) and (b). We usually find many Type (a) behaviors among just-as-processed devices. If the break of SLM operation is explained only by $\Delta \alpha L = 0$, all the dP/dI curves must be by Type (b), in which the curve is continuous at the SLM break point. This is because α_{th} does not change

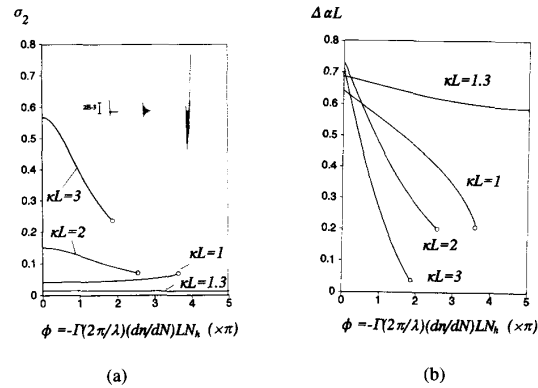


Fig. 4. Results of self-consistent iterative mode calculations using F -matrix as a function of normalized N_h . (a) σ_2 for the $\lambda/4$ -phase-shift DFB lasers with $\kappa L = 1, 1.3, 2$ and 3 . Examples of the traces of the iterative calculations at three points on the curve for $\kappa L = 2$ are inserted. (b) $\Delta \alpha L$ for the $\lambda/4$ -phase-shift DFB lasers with $\kappa L = 1, 1.3, 2$ and 3 .

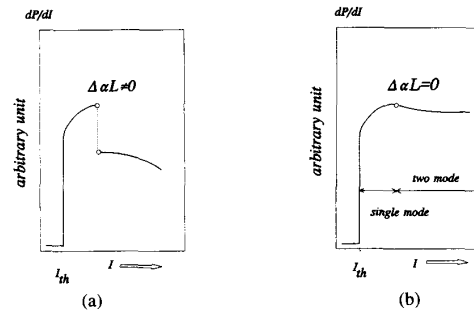


Fig. 5. Schematics of two types of dP/dI curves.

here. The existence of lasers with a discontinuous dP/dI curve as Type (a) proves that the condition of $\Delta \alpha L = 0$ is not always the cause of the SLM breaks.

B. dc Analysis

Figure 6(a), (b), (c), and (d) show dc-operated I - P and dP/dI curves for the $\lambda/4$ phase shift DFB lasers with $\kappa L = 3, 2, 1.3$, and 1 , respectively. The calculation step is 1 mA. Small fluctuations in dP/dI curves are due to calculation errors.

It is obvious that DFB lasers with larger $\sigma_{2,3,4}$ values tend to show a stronger nonlinearity in their I - P curve at lower currents.

C. Large-Signal Response

The pulse responses of output power and the lasing wavelength (chirp) biased at $I = I_{th} + 5$ mA are illustrated in Fig. 7(a) for $\kappa L = 1.3$ and 3 . A $<1010>$ bit sequence (2.4 Gbps) with rise and fall times of 50 ps and a height of 30 mA is applied. The calculations are carried out by a time interval of 1 ps.

The first spike of the relaxation oscillation is obviously suppressed when spatial holeburning is strong ($\kappa L = 3$), as is pointed out elsewhere [3]. The chirp after the first spike of

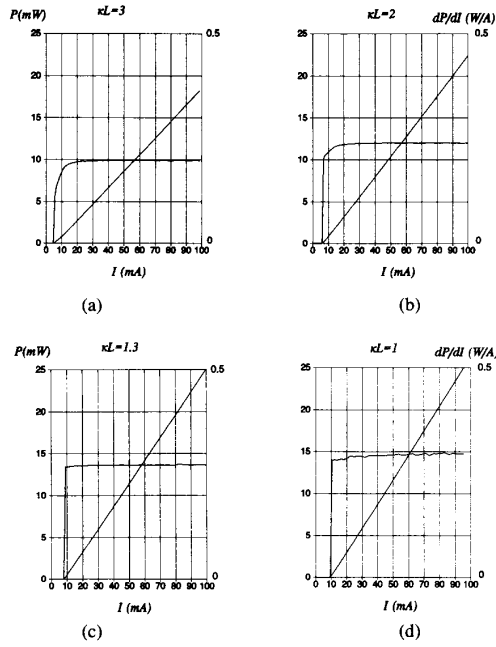


Fig. 6. Calculated I - P and dP/dI curves for $\kappa L = 3$ (a), 2 (b), 1.3 (c), 1 (d).

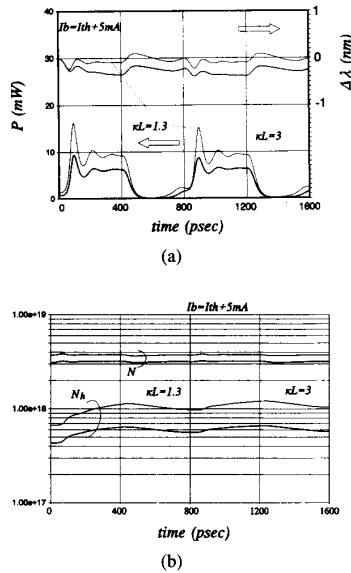


Fig. 7. Pulse responses biased at $I = I_{th} + 5$ mA for $\kappa L = 1.3$ and 3. (a) Output power and chirp. (b) N and N_h . (A series of current pulses of 2.4 Gbps (1010) pattern with rise and fall times of 50 ps, and a height of 30 mA is applied.)

the relaxation oscillation depends on the effect of holeburning on δ .

Fig. 7(b) shows the pulse responses of the average carrier density N and the deviation component N_h . The response of N_h is slower than N , giving rise to pattern effect more eminently.

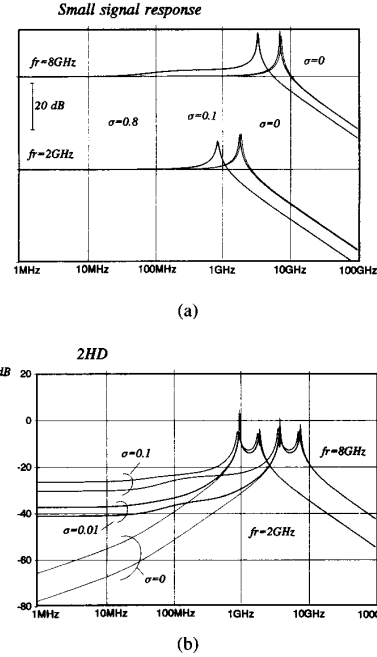


Fig. 8. Simplified small signal AM (a) and 2HD (b) responses calculated using (36)–(40), as a parameter of σ .

D. Simplified Small-Signal Analysis, Including 2HD

Before the AM analysis using (29) and (30), the estimations using the simplified formulas, (36)–(40) are shown in this subsection.

The calculated results are depicted in Fig. 8

(a) and (b) in comparison with the case in the absence of spatial holeburning ($\sigma = 0$). We assume $\tau_n = 3$ ns, and a modulation index of 4% for 2HD calculations. The simplified formulas are independent of DFB laser parameters except σ . The calculations are carried out for $\sigma = 0, 0.1$, and 0.8 in the first-order signal analysis, and for $\sigma = 0, 0.01$, and 0.1 in the 2HD analysis. Bias conditions can be normalized by the resonance frequency in the case of $\sigma = 0$. The bias conditions of $f_r = 2$ GHz and 8 GHz are used.

The AM responses shown in Fig. 8(a) give us information on the effect of the nonuniformity itself. The resonance frequencies of the lasers with relatively weak nonuniformity ($\sigma = 0.1$) are slightly lower than in the case of $\sigma = 0$. However, the devices with strong nonuniformity ($\sigma = 0.8$) has much lower resonance frequency. That is, very strong nonuniformity is found to reduce the resonance

The responses illustrated in Fig. 8(b) are useful to design DFB lasers for analog applications. The 2HD values are much worse than the ideal case ($\sigma = 0$) at lower frequencies even if they have relatively small nonuniformity ($\sigma = 0.1$). Even the lasers with extremely small nonuniformity ($\sigma = 0.01$) have worse 2HD values than in the ideal case at low frequencies. At higher frequencies, the resonance of relaxation oscillation dominates the 2HD behavior. It should be noted that the 2HD values are very sensitive to very small amounts of nonuniformity.

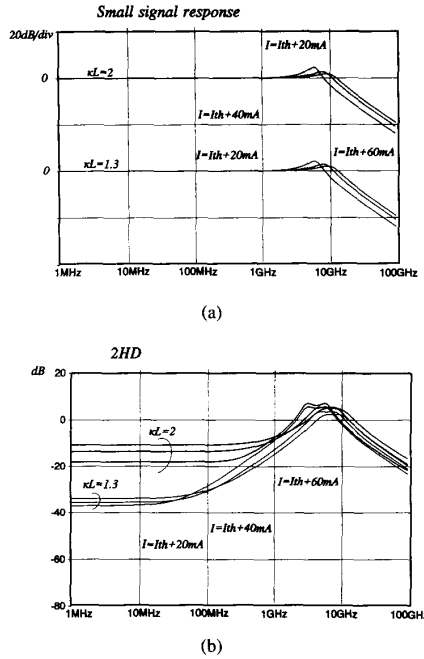


Fig. 9. Calculated small-signal AM responses (a), and 2HD responses (b), at bias currents of $I_{th} + 20$ mA, $I_{th} + 40$ mA, $I_{th} + 60$ mA for the $\lambda/4$ -phase-shift DFB lasers with $\kappa L = 1.3$ and 2.

E. Small-Signal Analysis, Including 2HD

The AM responses including 2HD in the absence of the parasitics at bias currents of $I_{th} + 20$ mA, $+40$ mA, and $+60$ mA are calculated using (29) and (30). The calculation is carried out at 500 frequency points from 1 MHz to 100 GHz. The $\lambda/4$ shift DFB lasers with $\kappa L = 1.3$, and 2 are assumed. The first-order AM signal and 2HD responses are shown in Fig. 9(a) and (b), respectively. The modulation index M is assumed to be 4%.

The difference between the AM responses for $\kappa L = 1.3$ and 2 is very small. This is because the σ_2 value for $\kappa L = 2$ becomes less than 0.1 after strong holeburning just above threshold, reducing the spatial nonuniformity. At bias current larger than $I_{th} + 20$ mA, holeburning is considered to affect the AM response not so seriously. In the plot of the AM responses as a parameter of I/I_{th} as in [8], the normalization factor I_{th} is affected by κL values. Plots as a parameter of applied current I as in Fig. 9 are thought to be more practical when comparing the AM responses of the various DFB laser structures, which have different κL values.

The difference in the 2HD responses between $\kappa L = 1.3$ and 2 is significantly large as predicted in the previous subsection. The 2HD values become worse as bias current increases at lower frequencies, whereas vice versa at higher frequencies. This is because the specific values of r -parameters and N_h greatly affect the 2HD behavior at lower frequencies although the resonance frequency dominates it at higher frequencies.

F. Accuracy and Discussion

The method presented here utilizes the approximations as explained in Section II.

A major one is the assumption that the photon and carrier density profiles along z , $S(z)$ and $N(z)$, have a common distribution function, $f(z)$. Some of the errors caused by this assumption are related to the nonlinearity in the rate equation system. For example, the profile $f(z)$ will be slightly changed because the region with larger photon densities is more affected by the nonlinear gain ϵ . Just above the threshold, the spontaneous emission and/or nonradiative emission will have a greater influence on the region with smaller photon densities. From this viewpoint, this method may be less rigorous particularly for the following cases:

1) Large-signal pulse response biased at currents below threshold.

2) High-power dc behavior for bias currents above 100 mA.

However, it will give us good approximation under practical operation conditions ($I_{th} < I < 100$ mA).

Regarding the mode-instability problem proposed in this paper, it will be also interesting to analyze it using a large-scale model. The experimental fact as shown in Fig. 5 suggests that the mode-instability occurs much before the next mode has the same gain as the lasing mode. In the calculation in [1], Soda *et al.* also observed similar numerical instability [18]. Lowery *et al.* have already shown the unstable oscillations in their calculated pulse responses, although they concluded that it was a result of beating between the two modes [10]. We should anyway pay much more attentions to such an instability problem.

The AM analysis is considered to give us good approximations. In fact, the 2HD responses in [19] are similar to our results. In particular, the calculated results in Fig. 9 in [19] almost agree to those in our Fig. 9, except for some differences such as the dB notation. (They utilize dBc.) The approximation used for 2HD analysis in [19] is quite similar to the formulation presented here. Although the formulation in [19] is valid for the limited case of small-signal analysis, our method can be applied even to the large-signal analysis. The feature of separating two equation systems facilitates analyzing types of DFB laser structures other than phase-shift DFB lasers with ideally zero-reflectivity facets.

The calculation errors also depend on the number of segments divided along the cavity, iteration steps, convergence criteria, and time intervals, etc. For example, we have confirmed that the calculation errors sufficiently decrease if the number of segments is larger than 30.

IV. SUMMARY

We propose a simple model for analyzing the dynamic behavior of DFB lasers with spatial holeburning.

We assume a z -dependent nonuniformity function common to both photon and carrier densities. On this assumption, we find that a self-consistent mode analysis can be performed using only the coupled-mode equations. In this analysis, we suggest that the mode instability (implying a SLM operation breakpoint) will occur before the next mode has the same gain as the lasing mode. The calculated results of the self-consistent mode analysis are applicable to dynamic analysis based on the

rate-equation formulation. As a result, we can calculate the two equation systems as if they were independent of each other.

The dynamic analysis, including large-signal responses, using only three rate equations greatly simplifies the calculations of the dynamic behaviors of DFB lasers.

Another important conclusion is that the existence of nonuniformity itself greatly affects the 2HD responses of DFB lasers even if their I - P characteristic is almost linear.

The calculations are carried out for $1.3\text{-}\mu\text{m}$ -region bulk $\lambda/4$ phase-shift DFB lasers with zero reflectivity facets. The calculated results well simulate the responses of the actual devices.

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